Future Makers
Research Community

Small Language Model Fine-Tuning

A key enabler to sovereign & efficient Agentic AI

24th April 2026

Marius Kiskemper

Atos

Personal introduction

Future Makers
Research Community

Marius Kiskemper

- Lead Data Scientist
- Data & AI Business Line
- Germany (Paderborn)
- B.Sc. Data Science
- M.Sc. Artificial Intelligence
- Focus on Sovereign Agentic AI & Language Model Fine-Tuning
- Lecturer at DHBW Mannheim

Atos

Contents

01

Motivation

Future Makers
Research Community

Atos

Motivation

SLMs vs. LLMs

98,3%

30x

80,6%

€20M

Motivation

Hardware requirements

Future Makers
Research Community

98,3%

… GPU cost reduction for inference

70B model:

- requires 140GB of vRAM
- Requires 2 NVIDIA H100 ($30k each)

→ $60k

7B model:

- requires 12GB of vRAM
- Requires 1 consumer-grade NVIDIA RTX ($1k)

→ $1k

Atos

Future Makers
Research Community

Motivation

SLMs vs. LLMs

98,3%

GPU cost reduction for inference

30x

80,6%

€20M

Atos

Motivation

Inference efficiency

30x

… more efficient in latency and energy consumption

Reducing:

- Latency
- Energy consumption
- FLOPs
- Infrastructure maintenance
- Neuron overhead

Further Information: Small Language Models are the Future of Agentic AI (NVIDIA)

Motivation

SLMs vs. LLMs

Future Makers Research Community

98,3%

GPU cost reduction for inference

30x

more efficient in latency and energy consumption

80,6%

€20M

Atos

Motivation

Fine-Tuning improves performance

Future Makers
Research Community

80,6%

… of tasks are performed better by fine-tuned SLMs, compared to GPT4

- Across 31 representative benchmark tests
  → In 25 benchmarks, the best fine-tuned model outperformed GPT4
- 72,2% of all fine-tuned models outperformed GPT4 (including 2B models)
- Narrow and domain-specific tasks are where specialized SLMs excel

Atos

Motivation

SLMs vs. LLMs

98,3%

GPU cost reduction for inference

30x

more efficient in latency and energy consumption

80,6%

of tasks are performed better by fine-tuned SLMs

€20M

Motivation

Data Sovereignty

€20M

… fine when causing a serious GDPR infringement

Infeasible to host a LLM sovereign and locally

→ Requires third-party API / infrastructure

→ Introduces risk of data breaches / GDPR exposure (especially when agents can extract data autonomously)

SLMs eliminate this risk by being easily locally hosted

Motivation

SLMs vs. LLMs

98,3%

GPU cost reduction for inference

30x

more efficient in latency and energy consumption

80,6%

of tasks are performed better by fine-tuned SLMs

€20M

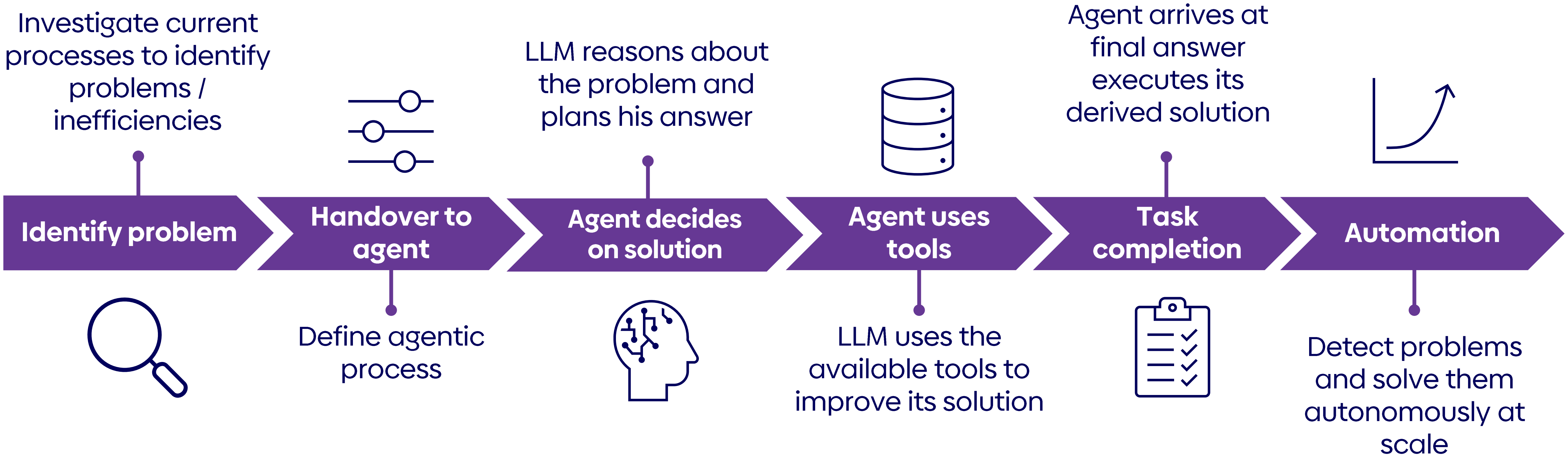
fine when causing a serious GDPR infringement

02

Solution Strategy

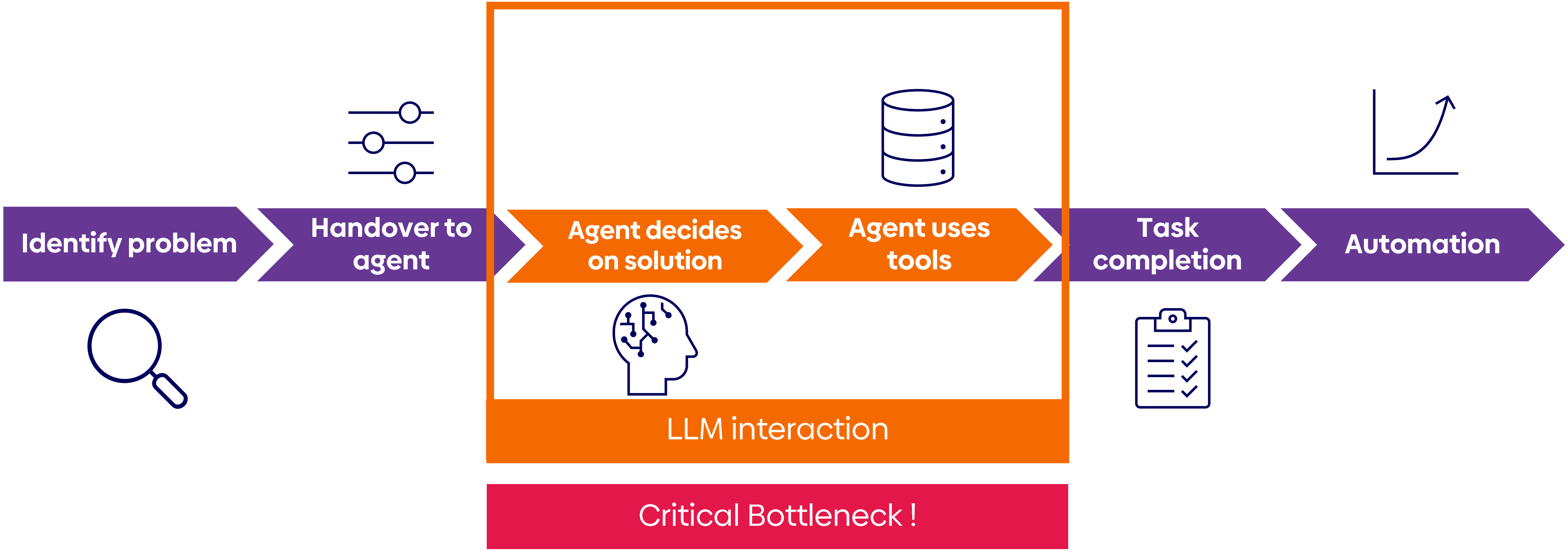
Solution Strategy

Typical Agentic AI solution process



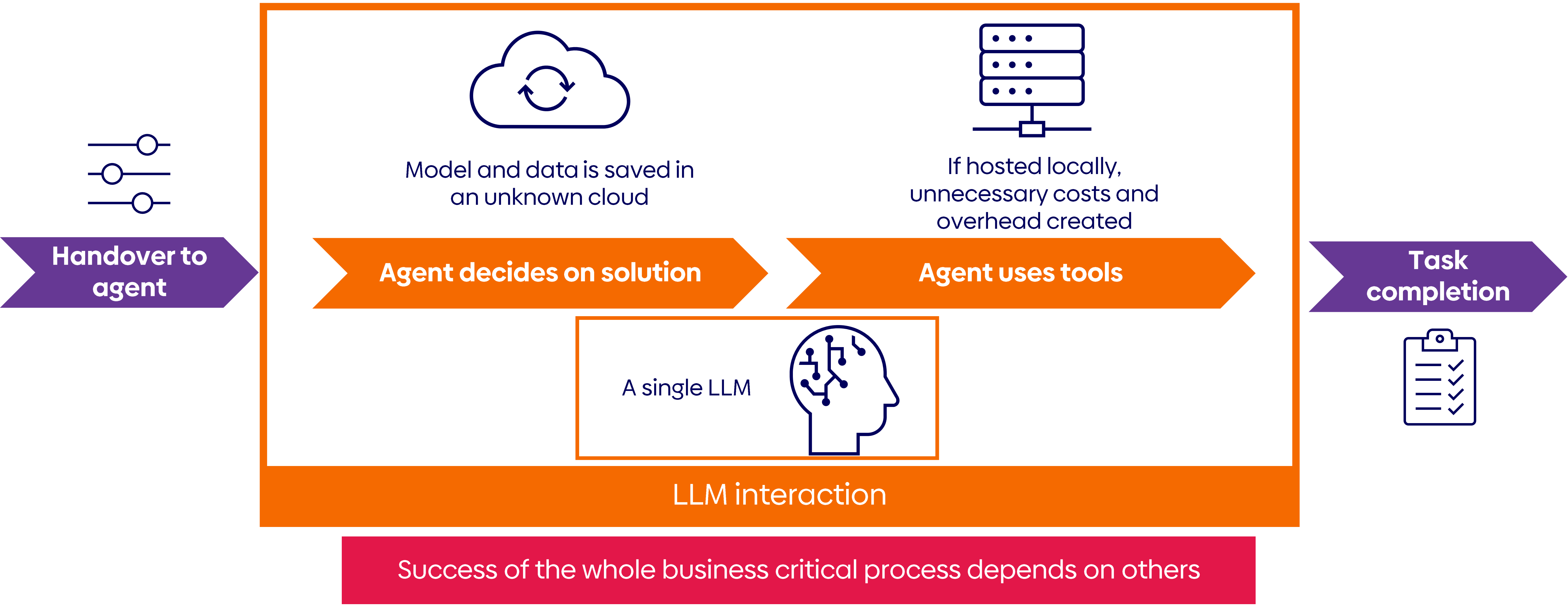
Solution Strategy

The risk of conventional agentic AI



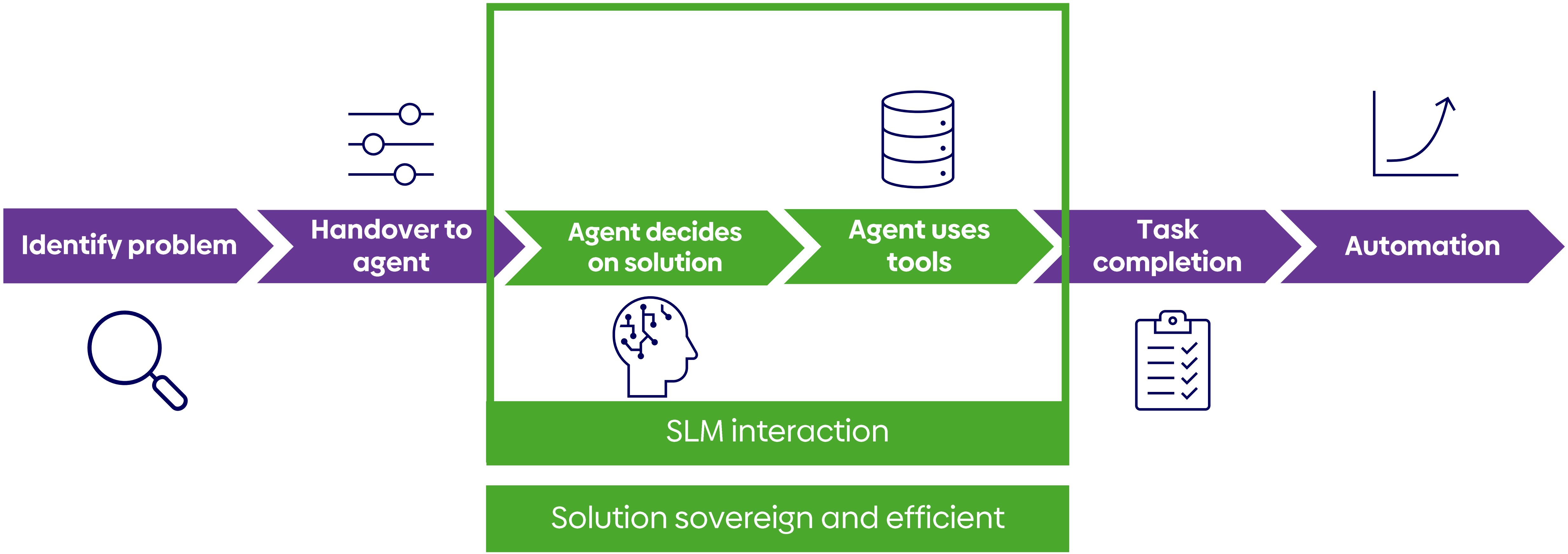
Solution strategy

The critical dependency



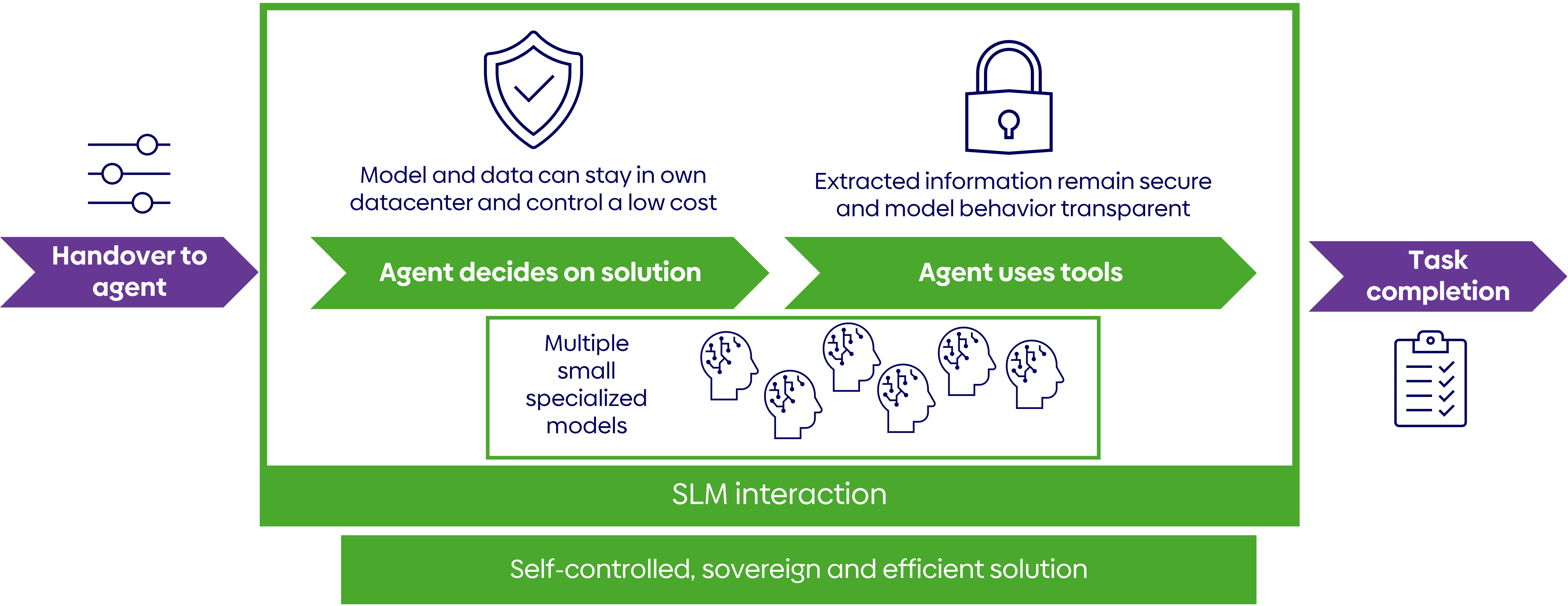
Solution Strategy

Integrating SLMs



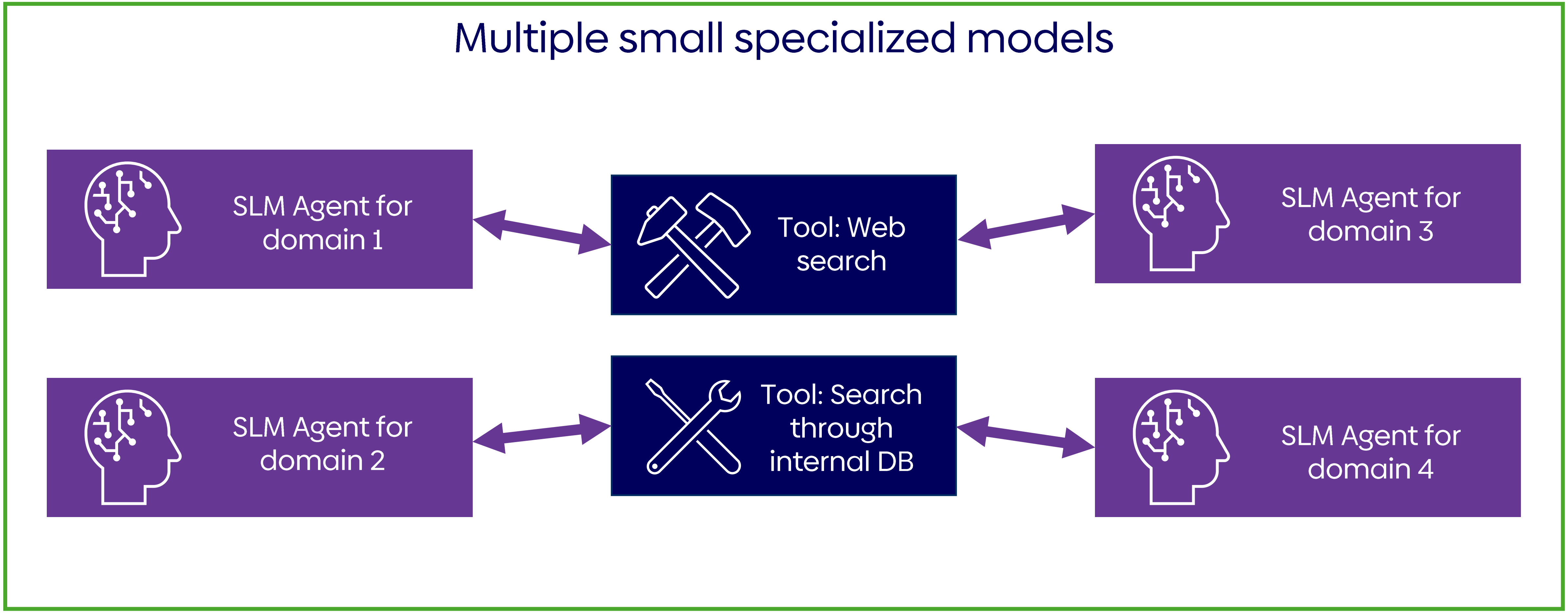
Solution strategy

Integrating SLMs



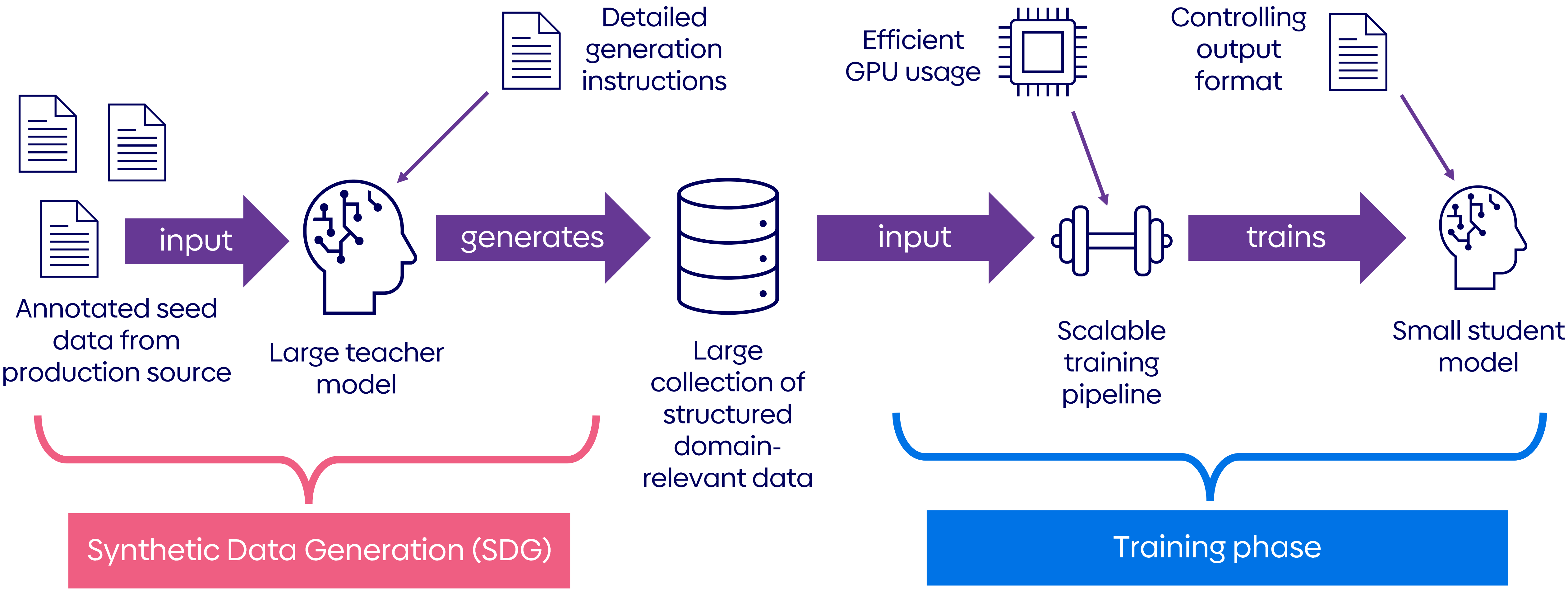
Solution strategy

Agentic orchestration



Solution strategy

The Fine-Tuning pipeline



03

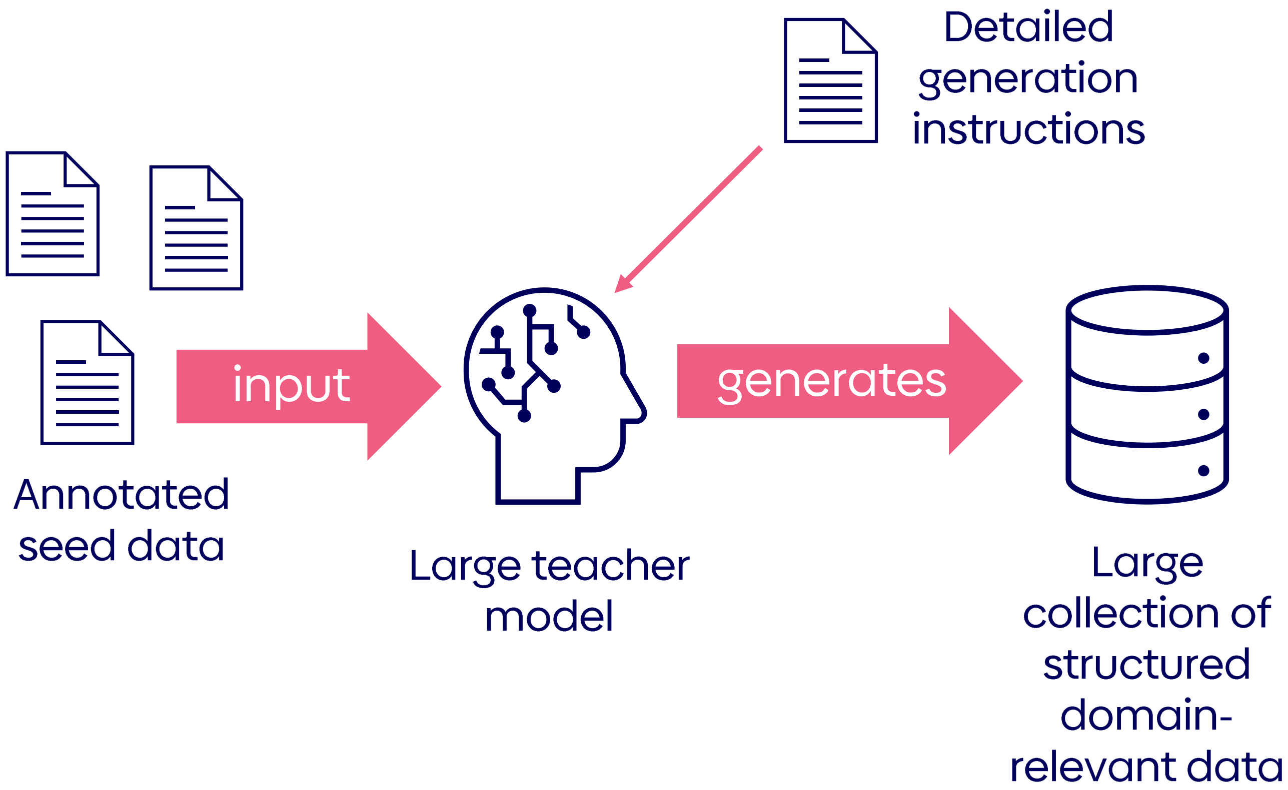
Technical Deep Dive

Future Makers
Research Community

Atos

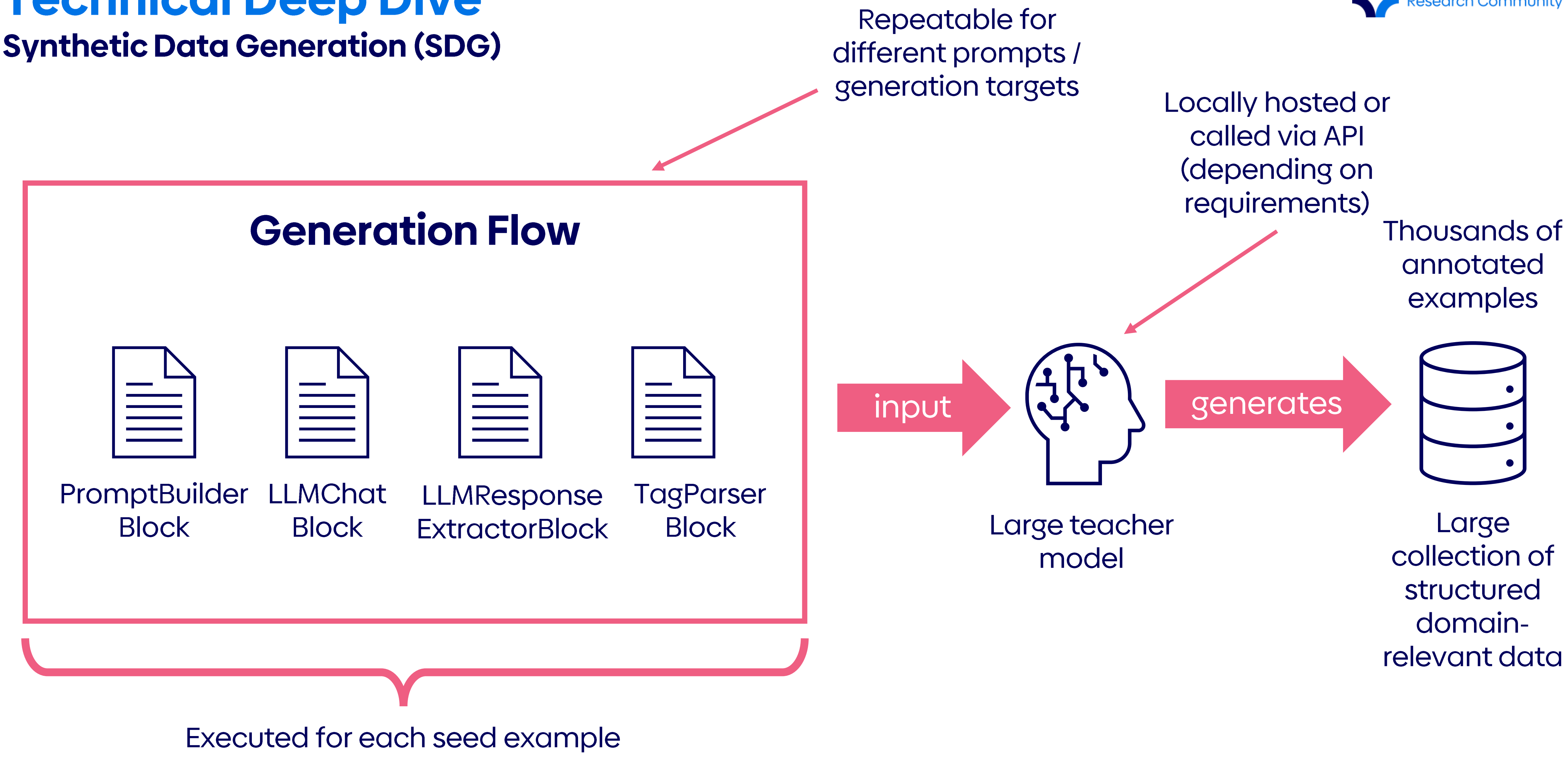
Technical Deep Dive

Synthetic Data Generation (SDG)



Technical Deep Dive

Synthetic Data Generation (SDG)



Technical Deep Dive

Infrastructure difficulties

Future Makers
Research Community

- CUDA and PyTorch as the center
- NVIDIA GB10 (with ARM CPU instead of x86 CPU)
  - Most regular packages built for x86
- CUDA version → CUDA 13 vs 12.x
- PyTorch typically only supports up to SM_90a (H100)
- Blackwell uses microarchiteture SM120
  - Streaming multiprocessor → actual computation architecture
  - SM Version: Compute capability (e.g. new instructions, new tensor mechanisms, …)
- Solution: NGC (NVIDIA GPU Cloud) PyTorch image
  - Compiles PyTorch differently (specifically for NVIDIA GPUs)
  - But this is newer than most training backends (unsloth, …)
- No vLLM image for CUDA 13 and SF120

→ No dealbreaker. But: Be aware ☺

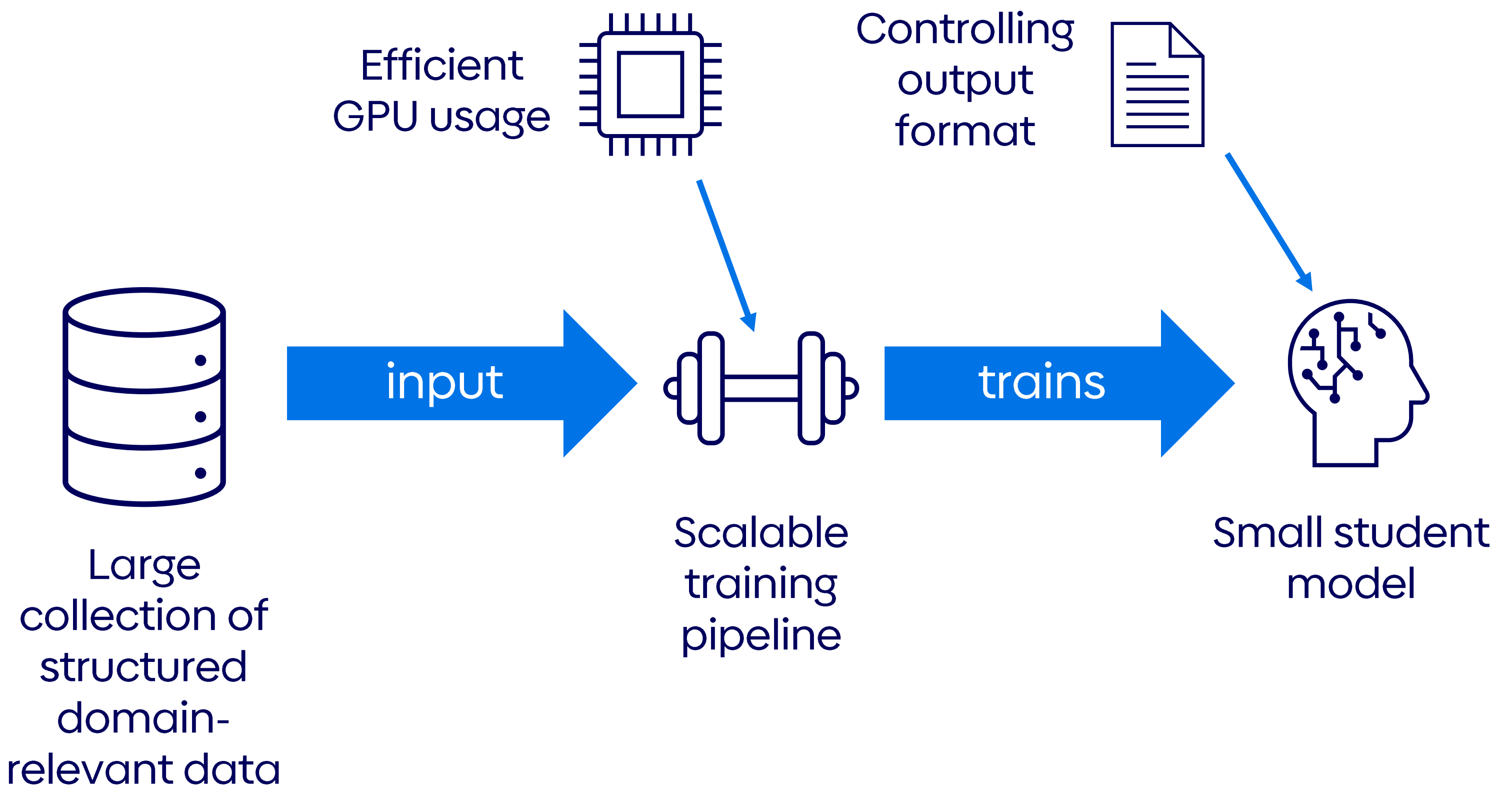
Future Makers
Research Community

Dive into the code: SDG

Atos

Technical Deep Dive

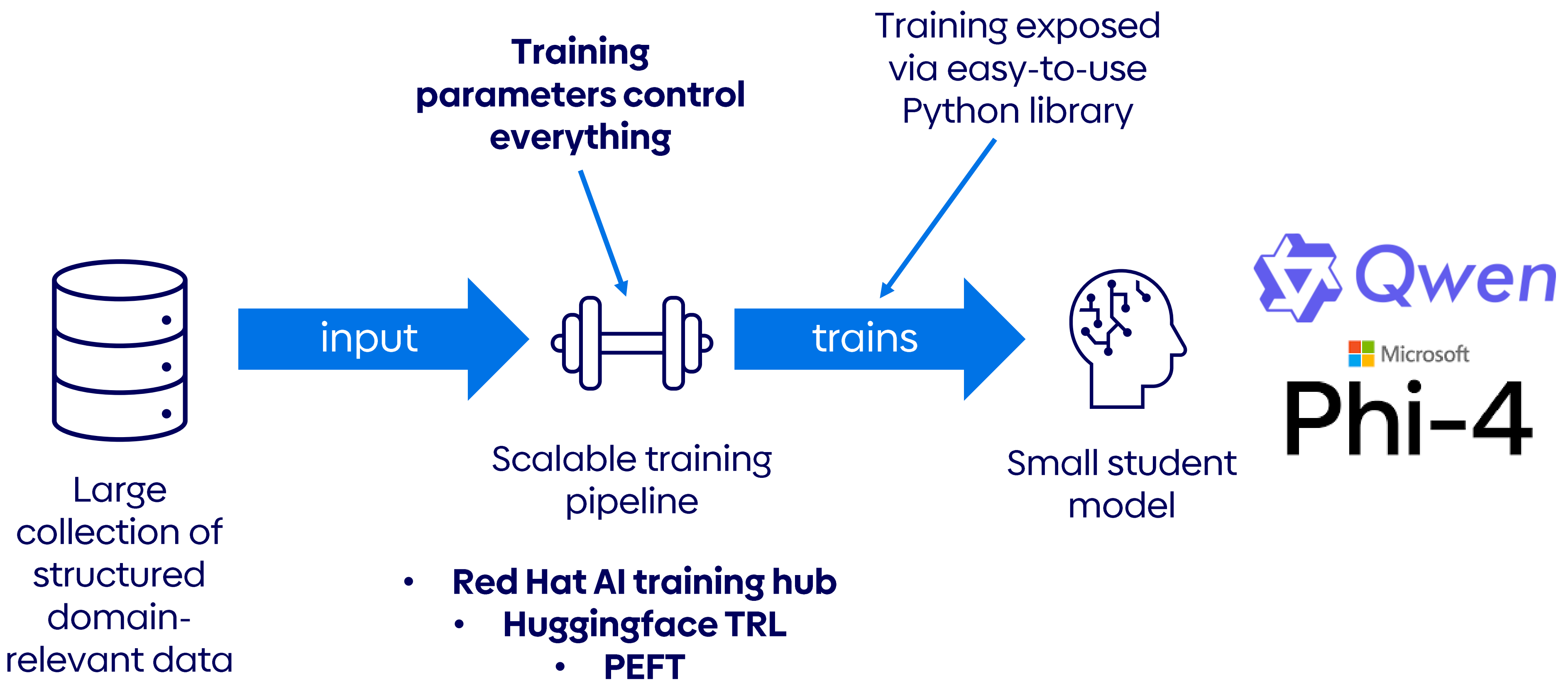
Training Phase



Technical Deep Dive

Training Phase

Future Makers
Research Community



Atos

Future Makers
Research Community

Dive into the code: Fine-Tuning

Atos

04

Results & Scaling it !

Results

Project use case example

Future Makers Research Community

„Liebe Leute die Polizisten brechen gerade ihr Eid und begehen Hochverrat schlagt sie mit alles was dir in die Hand kriegen könnt schlagt sie zu brei ACAB“

German user comments

Automated analysis


```
{
    "companies": [],
    "persons": [],
    "locations": [],
    "organizations": [],
    "groups": ["Polizisten"],
    "summary": "Der Nutzer beschuldigt Polizisten des Hochverrats
        und ruft dazu auf, sie mit Gewalt anzugreifen.",
    "language": "DE",
    "adjectives": [],
    "sentiment": "Negative",
    "tonality": ["inciting", "aggressive"],
    "spelling_errors": ["mit alles (mit allem)", "dir (ihr)",
        "kriegen (kriegen könnt)", "zu brei (zu Brei)"],
    "vulgarity": [],
    "slurs": ["ACAB"],
    "violence_triggers": ["schlagt sie", "schlagt sie zu brei"],
    "violence_detection": "true“
}
```

Results

Project use case example


```
{
    "companies": [],
    "persons": [],
    "locations": [],
    "organizations": [],
    "groups": ["Polizisten"],
    "summary": "Der Nutzer beschuldigt Polizisten des Hochverrats
        und ruft dazu auf, sie mit Gewalt anzugreifen.",
    "language": "DE",
    "adjectives": [],
    "sentiment": "Negative",
    "tonality": ["inciting", "aggressive"],
    "spelling_errors": ["mit alles (mit allem)", "dir (ihr)",
        "kriegen (kriegen könnt)", "zu brei (zu Brei)"],
    "vulgarity": [],
    "slurs": ["ACAB"],
    "violence_triggers": ["schlagt sie", "schlagt sie zu brei"],
    "violence_detection": "true“
}
```


Key challenges:

- **Correct label** classification
- Constantly creating output in a **valid JSON format**
- JSON Extraction correct content across 14 different categories and interpreting German social media data
- **Multilingual** complexity (German language data)
- Can’t be solved with more context → RAG not the answer

Results

Evaluation methodology

Future Makers
Research Community

„Liebe Leute die Polizisten brechen gerade ihr Eid und begehen Hochverrat schlagt sie mit alles was dir in die Hand kriegen könnt schlagt sie zu brei ACAB“

German user comments

Automated analysis

Model output

{
"companies": [],
"persons": [],
"locations": [],
"organizations": [],
"groups": ["Polizisten"],
"summary": "Der Nutzer beschuldigt Polizisten des Hochverrats und ruft dazu auf, sie mit Gewalt anzugreifen.",
"language": "DE",
"adjectives": [],
"sentiment": "Negative",
"tonality": ["inciting", "aggressive"],
"spelling_errors": ["mit alles (mit allem)", "dir (ihr)", "kriegen (kriegen könnt)", "zu brei (zu Brei)"],
"vulgarity": [],
"slurs": ["ACAB"],
"violence_triggers": ["schlagt sie", "schlagt sie zu brei"],
"violence_detection": "true“
}

Atos

Results

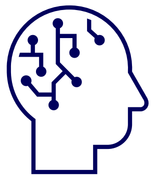
Evaluation methodology

Model output


```
{
	"companies": [],
	"persons": [],
	"locations": [],
	"organizations": [],
	"groups": ["Polizisten"],
	"summary": "Der Nutzer beschuldigt Polizisten des
	Hochverrats		und ruft dazu auf, sie mit Gewalt
	anzugreifen.",
	"language": "DE",
	"adjectives": [],
	"sentiment": "Negative",
	"tonality": ["inciting", "aggressive"],
	"spelling_errors": ["mit alles (mit allem)", "dir (ihr)",
		"kriegen (kriegen könnt)", "zu brei (zu Brei)"],
	"vulgarity": [],
	"slurs": ["ACAB"],
	"violence_triggers": ["schlagt sie", "schlagt sie zu brei"],
	"violence_detection": "true“
}
```


LLM-as-a-judge

Compares



Groundtruth


```
{
	"companies": [],
	"persons": [],
	"locations": [],
	"organizations": [],
	"groups": ["Polizisten"],
	"summary": "Der Nutzer beschuldigt Polizisten des
	Hochverrats		und ruft dazu auf, sie mit Gewalt
	anzugreifen.",
	"language": "DE",
	"adjectives": [],
	"sentiment": "Negative",
	"tonality": ["inciting", "aggressive"],
	"spelling_errors": ["mit alles (mit allem)", "dir (ihr)",
		"kriegen (kriegen könnt)", "zu brei (zu Brei)"],
	"vulgarity": [],
	"slurs": ["ACAB"],
	"violence_triggers": ["schlagt sie", "schlagt sie zu brei"],
	"violence_detection": "true“
}
```

Results

Evaluation metric

Future Makers Research Community

28 **Possible points for information extraction**
- Extracting all named entities, slurs, tonality, vulgarity, violence triggers, spelling errors
- 2 points for each correctly extracted feature

3 **Possible points for JSON correctness**
- JSON is machine readable without any error
- 3 points or 0 –> no in between scoring

3 **Possible points for correct label prediction**
- Whether the prediction matches the human gold standard
- 3 points or 0 -> no in between scoring

34 **Total points achievable per analyzed input**

Atos

Results

Model setup

Future Makers
Research Community

Baseline models

- GPT 4.1 (via API)
- Phi4-mini (3.5B)
- Qwen 2.5 (1.5B)
- Qwen 2.5 (7B)
- NuExtract (2B)
- NuExtract (8B)

Prompting Strategies

- Base prompt
  - One-Shot (Short description and no examples)
- Detailed prompt
  - Multi-Shot (multiple examples and very detailed task description)

- 6 base models x 2 prompt strategies

→ 12 base model outputs

Fine-Tuning variations

- SFT (Supervised Fine-Tuning)
  - 3 checkpoints (3 epochs)
- OSFT (Orthogonal Subspace Fine Tuning)
  - 3 checkpoints (3 epochs)
  - Two different parameter configs

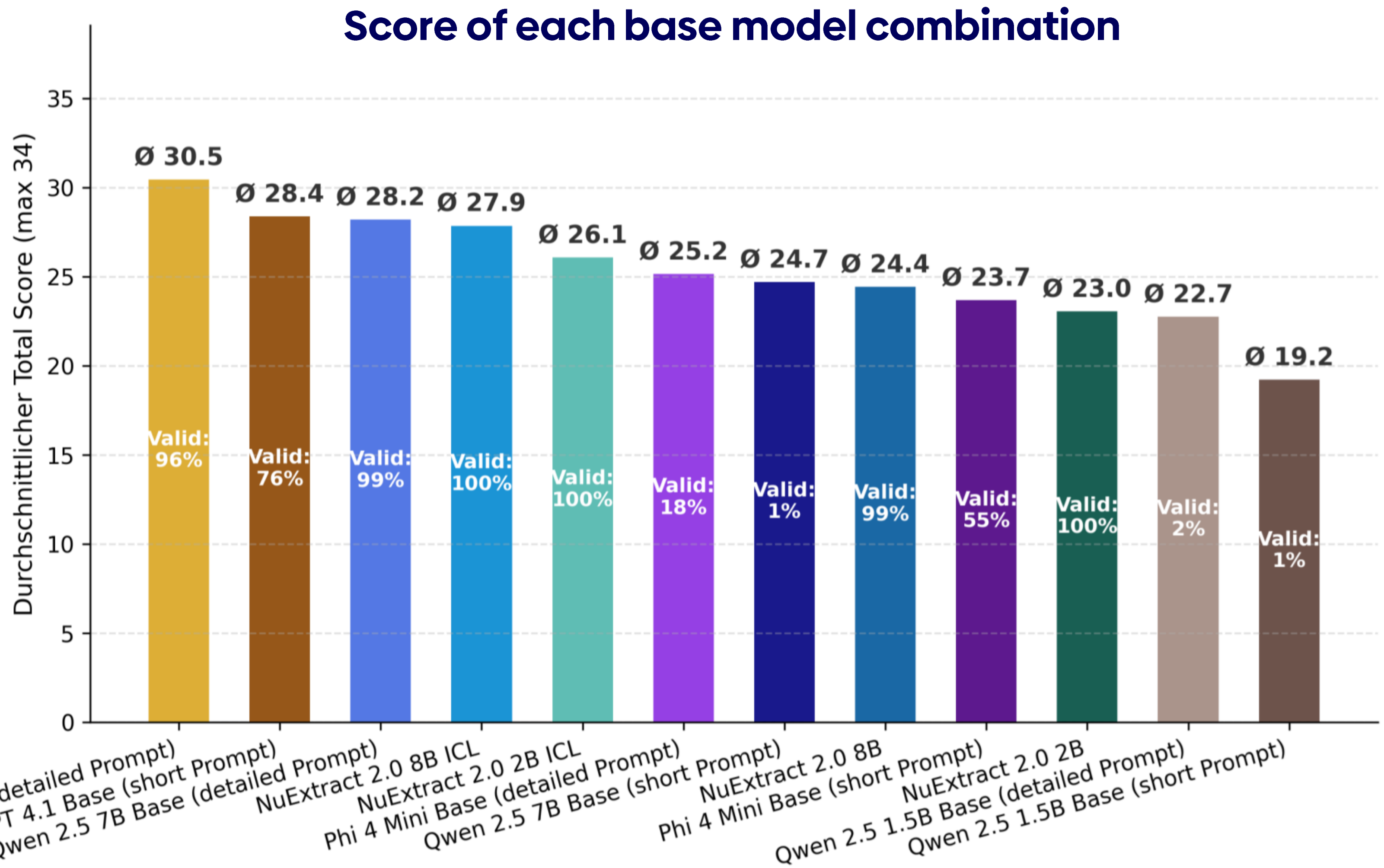
- 3 student models x 9 fine-tuning configurations

→ 27 fine-tuned model outputs

Atos

Results

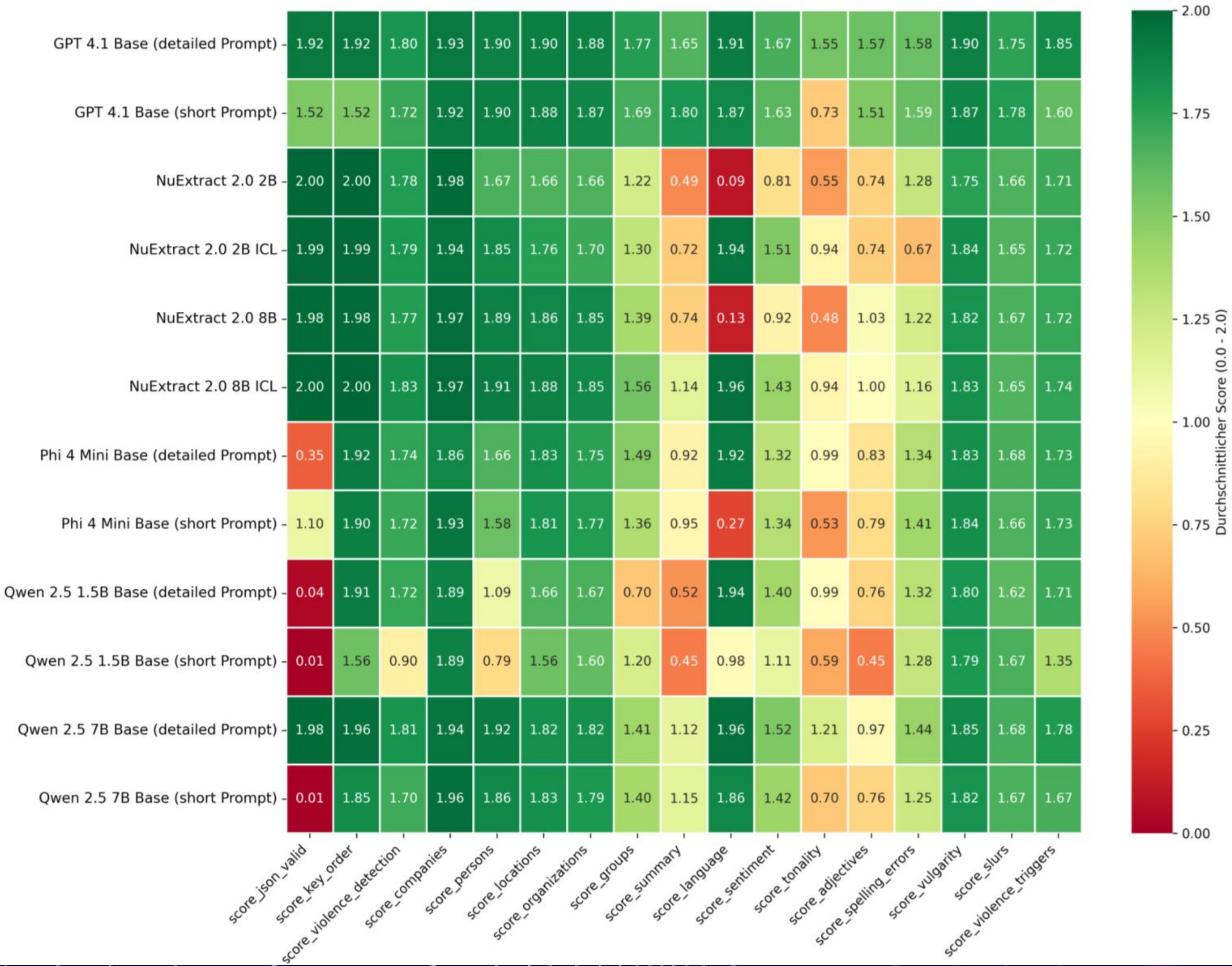
Base model performances



Results

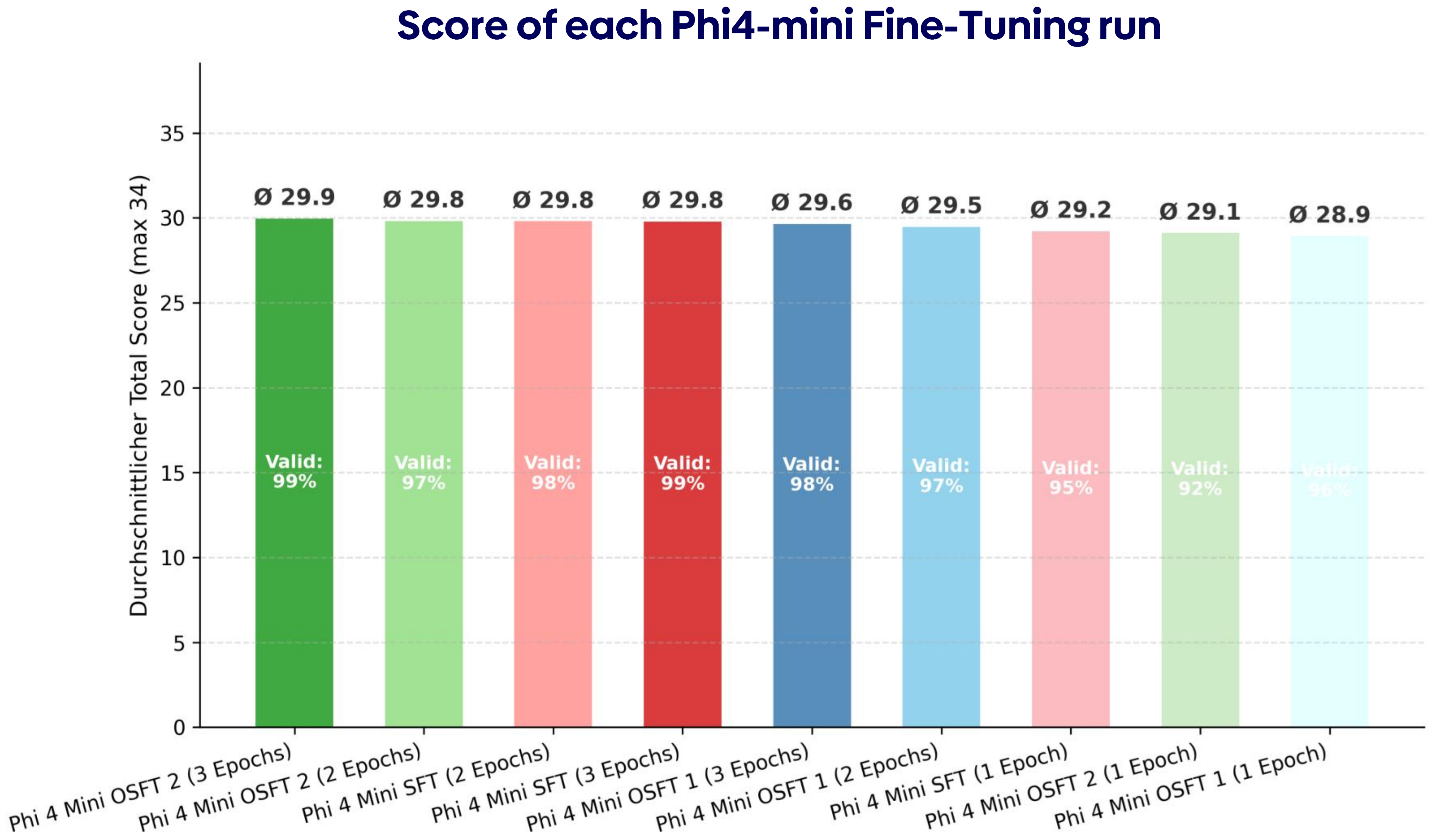
Base model performances

Base model scores across the different extraction targets



Results

Fine-Tuning performance: Phi4-mini

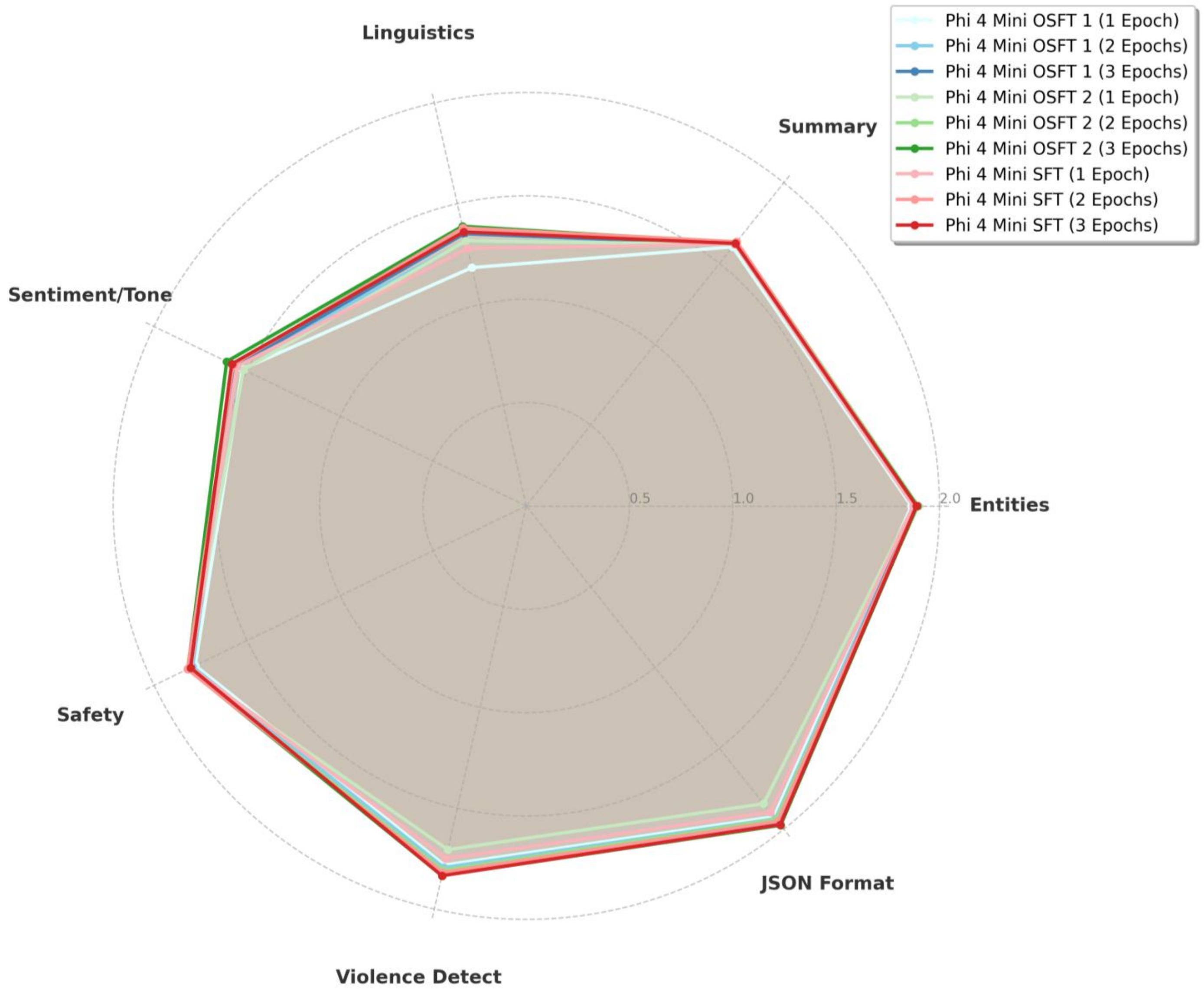


Results

Fine-Tuning performance: Phi4-mini

Future Makers
Research Community

Score of each Phi4-mini Fine-Tuning run across the different tasks

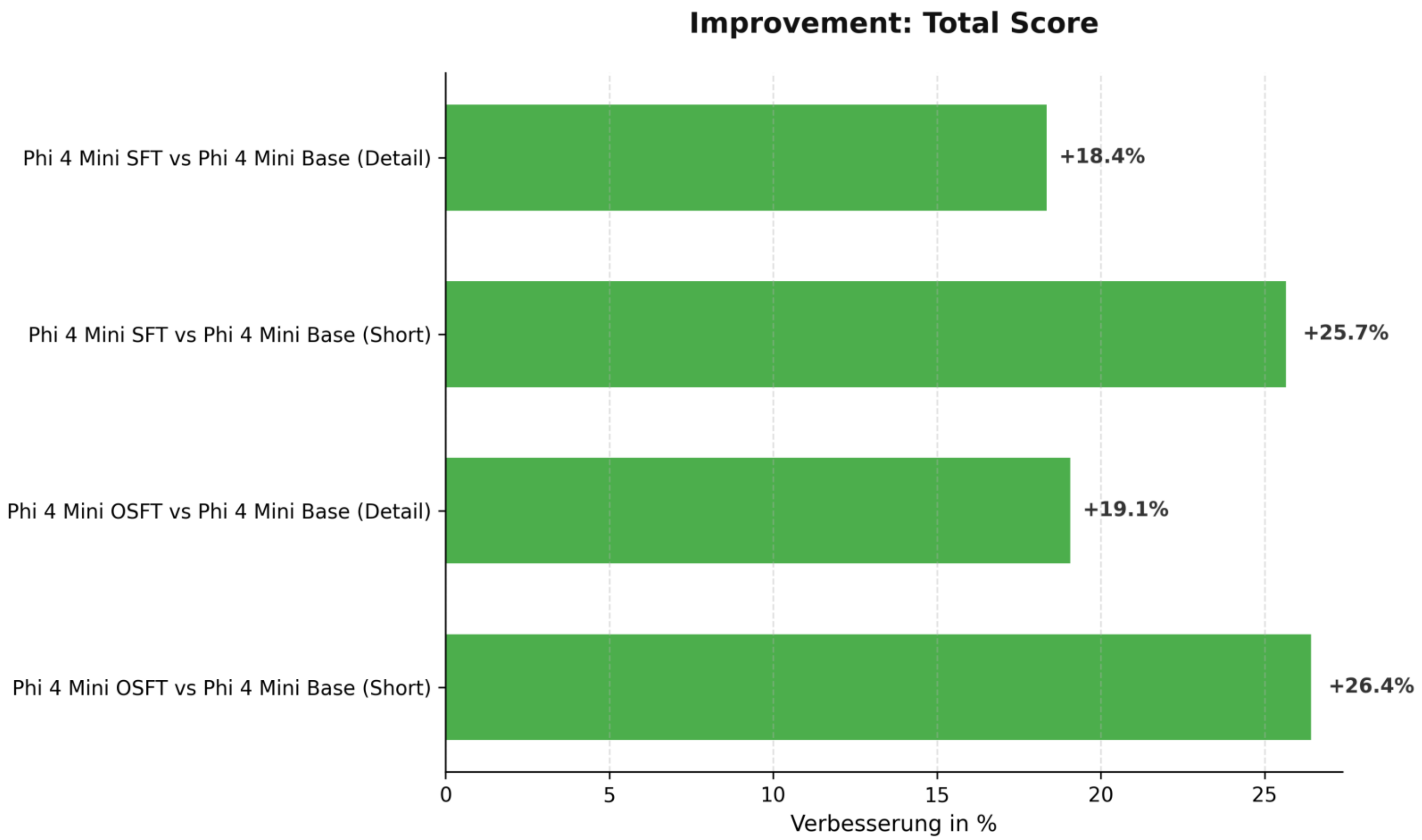


Atos

Results

Fine-Tuning performance: Phi4-mini

Total score improvements, compared to the Phi4 base models

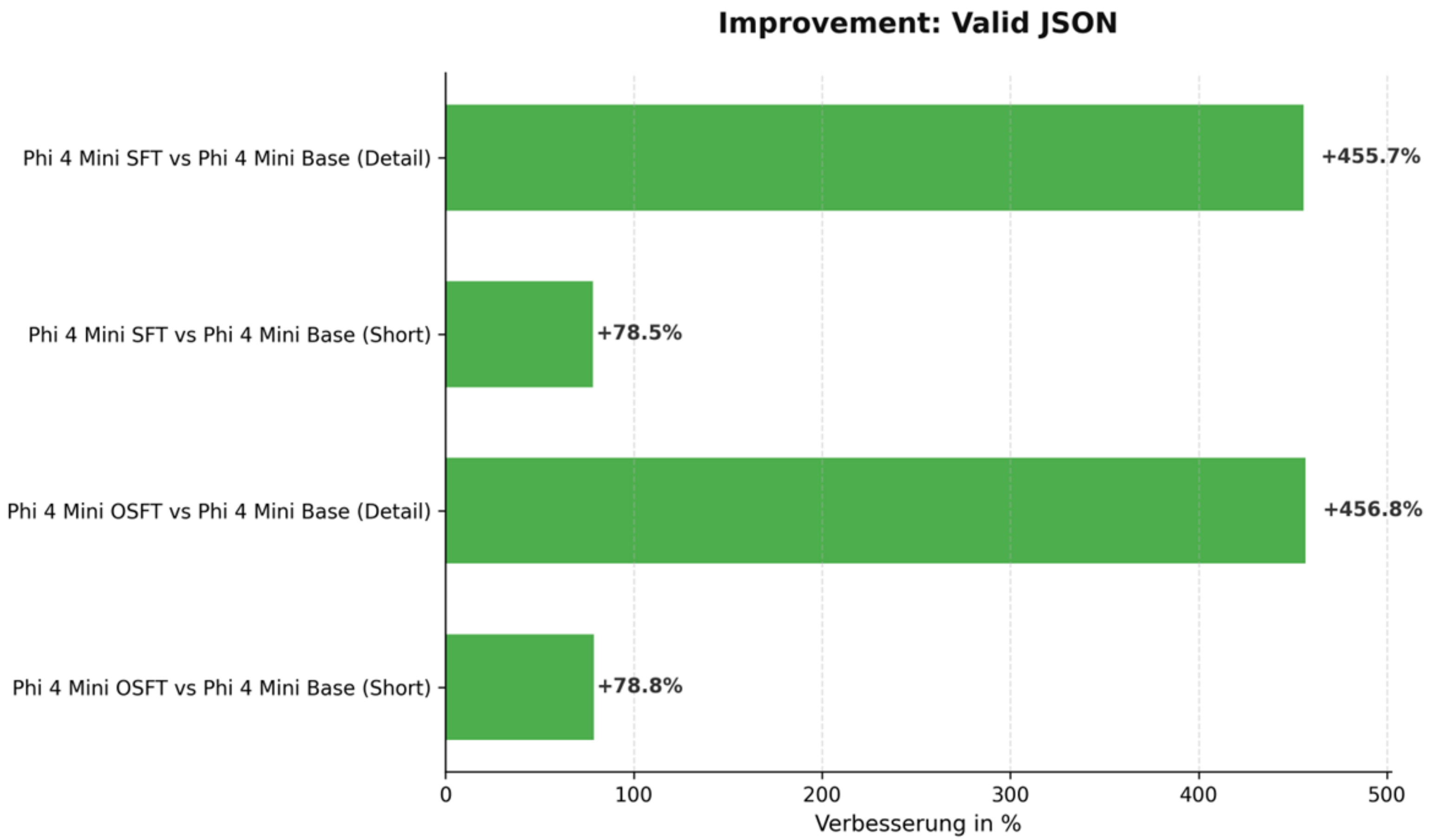


Results

Fine-Tuning performance: Phi4-mini

Future Makers Research Community

Valid JSON improvements, compared to the Phi4 base models



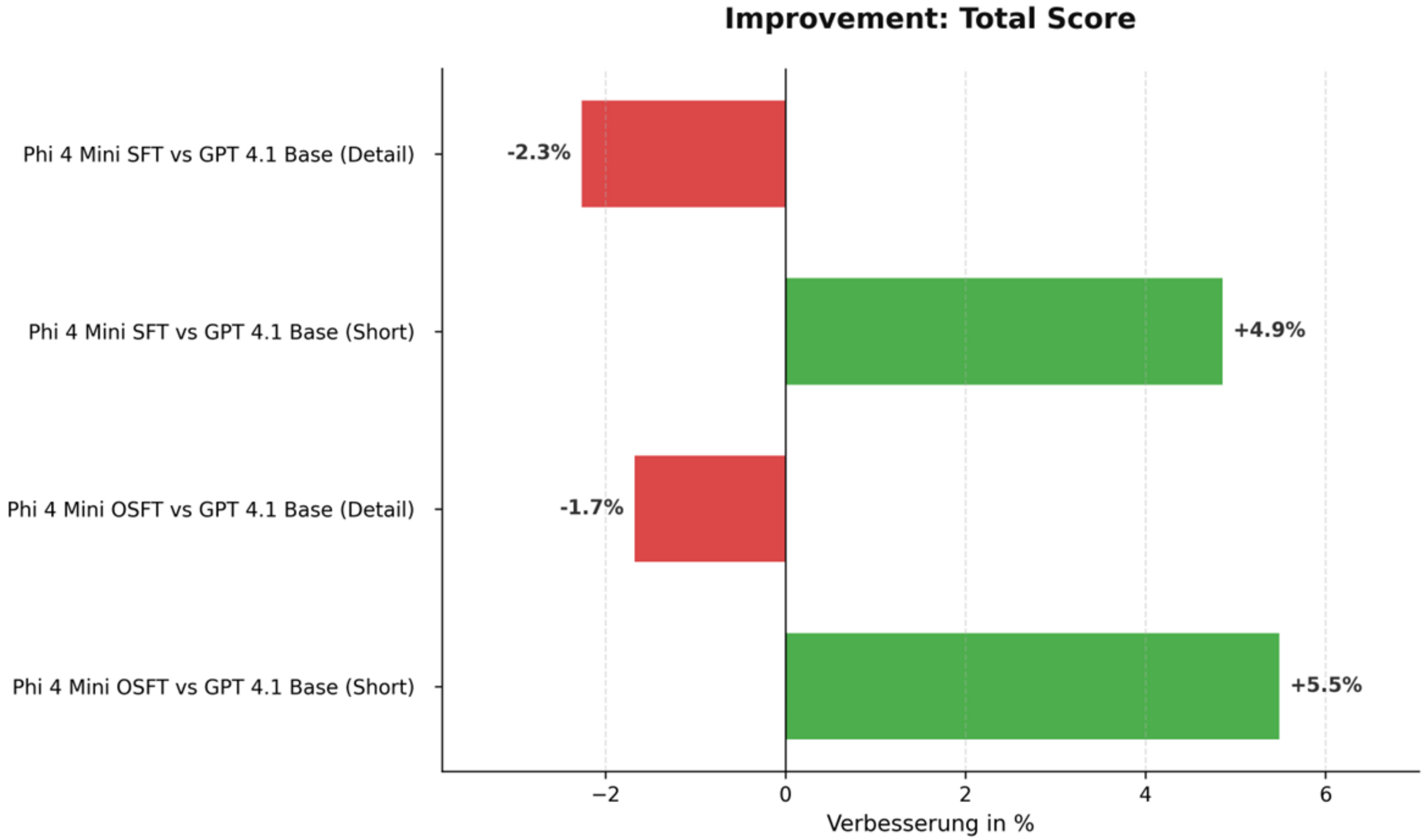
Atos

Results

Fine-Tuning performance: Phi4-mini vs. GPT4.1

Future Makers
Research Community

Total score improvements, compared to the GPT4.1 model

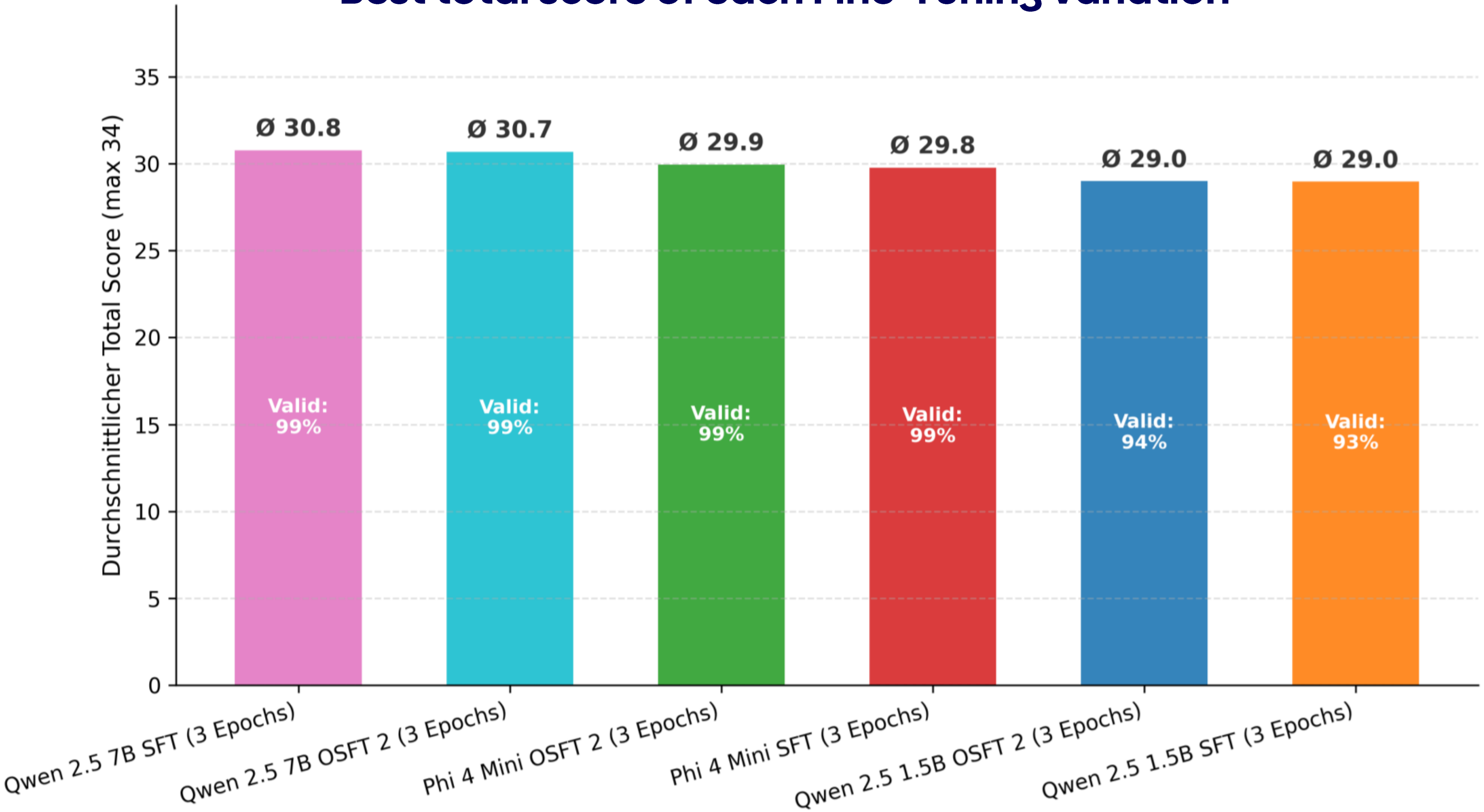


Atos

Results

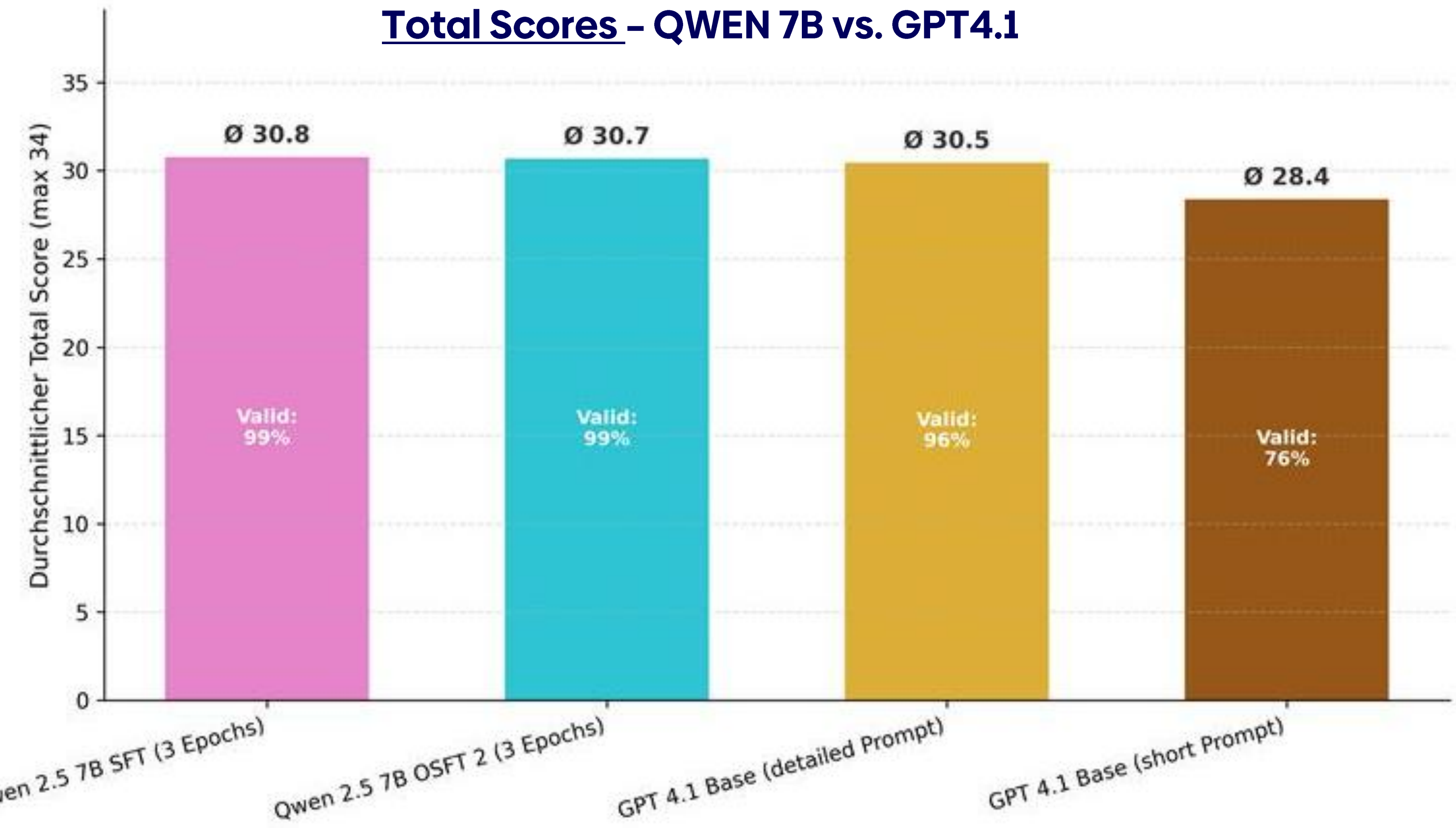
Overall best fine-tuned models

Best total score of each Fine-Tuning variation



Results

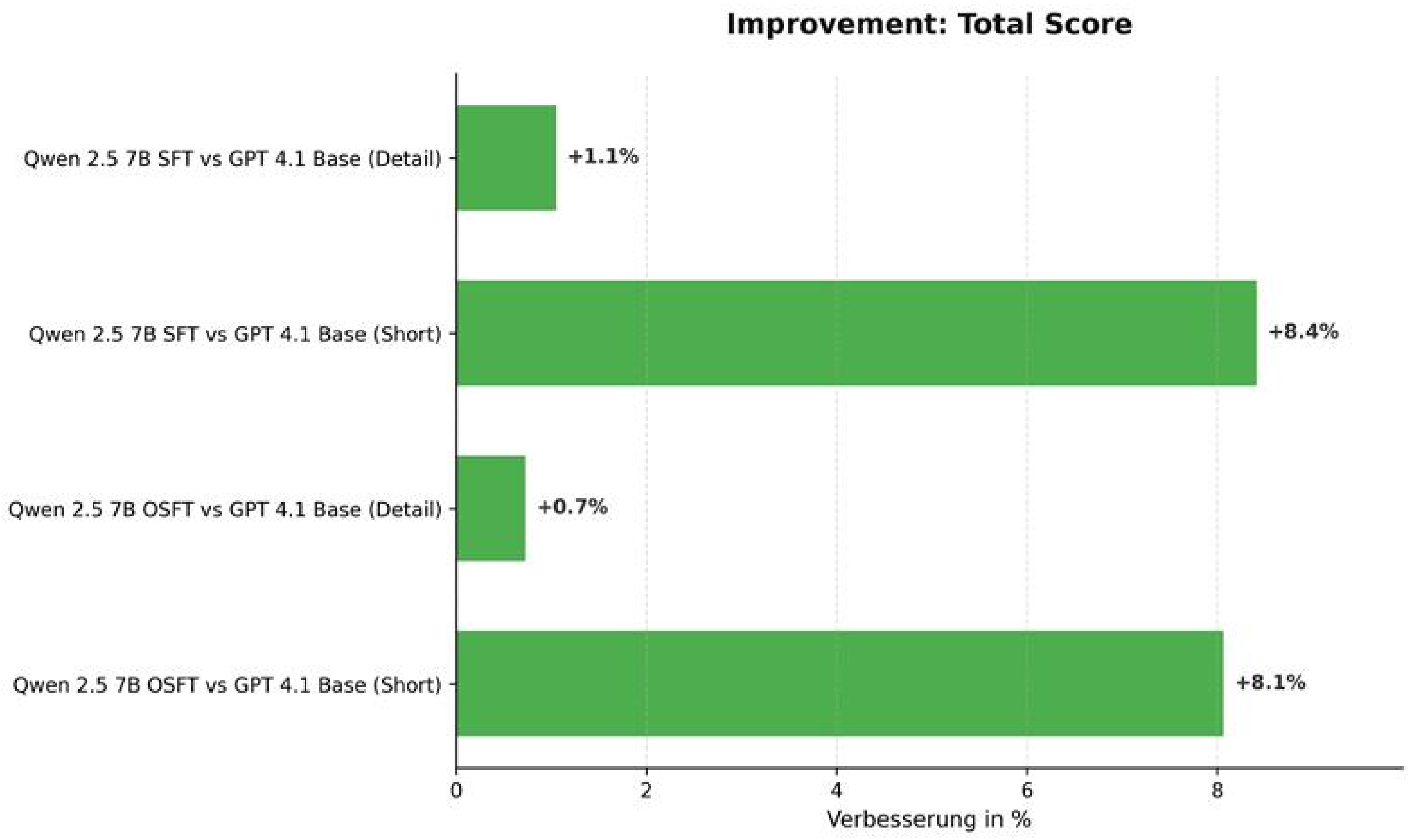
Finetuned QWEN 7B outperforms GPT4.1



Results

Finetuned QWEN 7B outperforms GPT4.1

Total Score improvements – QWEN 7B vs. GPT4.1



Scaling it !

Bringing fine-tuned SLMs to productive Agentic AI

- Fine-tuned SLMs outperform generalistic LLMs on narrow and domain-specific tasks
- SLMs are significantly cheaper for inference and can be deployed completely sovereign
- The shown approach can be scaled to all kinds of customer use cases
- Using SLMs in agentic AI improves control over model behavior and allows to choose the right tool for the given problem


revex


Future Makers
Research Community

Q & A

Atos

Sources

Sources

Contents

- GPU vRAM costs: Best GPUs for LLM inference in 2025 | WhiteFiber
- Inference efficiency: Small Language Models are the Future of Agentic AI
- Fine-Tuning vs. GPT4: 2405.00732
- Neuron overhead: https://aclanthology.org/2024.acl-long.678.pdf
- GDPR fines: What are the GDPR Fines? - GDPR.eu

Future Makers
Research Community

Thank you

For more information please contact:

Marius.Kiskemper@atos.net

Atos